R99 | Global balanced B versus row-softmax A

Same fresh selector + input-only32 early fusion; only column competition differs.

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	-0.012071	12491	12667	0.000097670	0.001615
VAL	-0.035802	677	706	0.000075197	0.000356
TEST	0.032099	706	680	0.000137774	0.001513

Split	Target correct	Rescued	Lost	New false	Recall (%)
TRAIN	436	111	187	114	0.5235
VAL	23	7	10	6	0.4973
TEST	21	5	10	8	0.4567

A row-softmax / B fixed64 Sinkhorn. Same raw8 attributes, input-only32 and projection seed990003.

Original CE .35 / comparison .35 / risk .02 / R-Drop .35, dropout .1. No new target/loss.

Original four instance layers, R88B physical TSP anchor, hand34/methodID, joint2, score MLP/maximin.

Representational accessibility, not an assignment solver, VRP bound or B-semantics repair.

R89B main / R84A secondary historical; no newly matched historical controls. Fixed target36.

ALL deltas equal-weight18 tasks; native FP64 winners, TRAIN SBS; no TEST/LIB model selection.

R99 | Historical VAL screens and hardware

Descriptive locked-model diagnostics, not checkpoint/model reselection.

Arm / reference	Top1 delta pp	Cost delta	Regret delta pp	Pass
R99/A / R89B	-0.154321	0.000601406	0.004483	False
R99/A / R84A	0.008642	0.000284212	0.001665	False
R99/B / R89B	-0.190123	0.000676602	0.004840	False
R99/B / R84A	-0.027160	0.000359409	0.002022	False

A: {"gpu_type": "3090"}

B: {"gpu_type": "3090"}

Disclosed hardware is not bitwise hardware matching. Development-exposed TEST.

R99 B vs A | A Input Transform / Inference Timing

First128 locked TEST input rows/task; synchronized GPU wall time, three repetitions; saved evidence.

Problem	Transform ms/row	Transform + selector ms/row	Cache-hit selector ms/row	Marginal residual
TSP	0.458086	0.858860	0.394064	1.15798
CVRP	0.596066	1.077541	0.478759	1.29677
ATSP	0.012177	0.084773	0.072415	5.3432
OVRP	0.031181	0.101733	0.070387	1.27029
VRPB	0.031275	0.101671	0.070216	1.48684
VRPL	0.031326	0.101197	0.069915	1.36033
VRPTW	0.031272	0.101556	0.070193	1.50045
OVRPTW	0.031680	0.104389	0.073580	1.44011
OVRPB	0.031635	0.102252	0.071098	1.75916
OVRPL	0.031301	0.103080	0.070466	1.26628
VRPBL	0.031294	0.102519	0.070309	1.28209
VRPBTW	0.031127	0.102628	0.070113	1.21418
VRPLTW	0.032597	0.105631	0.070552	1.36161
OVRPBL	0.031346	0.102194	0.070404	1.50991
OVRPBTW	0.031305	0.101744	0.071080	1.28365
OVRPLTW	0.031672	0.101782	0.071823	1.26635
VRPBLTW	0.031608	0.102305	0.070062	1.29172
OVRPBLTW	0.031136	0.102185	0.070369	1.38966

Feature preparation stage wall time: TRAIN/VAL 125.736s; TEST 9.839s (cache reuse allowed).

Original inputs already on GPU; online raw path has no relation_nodes/relation_node_mask.

Transform includes FP16 quantization. All18 tasks include ATSP; no solver decoder.

Excludes dataset I/O and host transfer. Cache-hit selector is not full inference or I/O-inclusive E2E.

Finite64 residuals are recorded, not proof of exact Sinkhorn convergence. Device: NVIDIA GeForce RTX 3090

R99 B vs A | B Input Transform / Inference Timing

First128 locked TEST input rows/task; synchronized GPU wall time, three repetitions; saved evidence.

Problem	Transform ms/row	Transform + selector ms/row	Cache-hit selector ms/row	Marginal residual
TSP	1.397523	1.800659	0.397854	0.0168889
CVRP	1.792300	2.282085	0.485822	0.0165278
ATSP	0.102403	0.176832	0.074030	0.0358955
OVRP	0.121215	0.192904	0.071828	0.0210742
VRPB	0.120090	0.192935	0.071511	0.0178392
VRPL	0.120463	0.192968	0.071357	0.0117588
VRPTW	0.120821	0.192403	0.071012	0.0236063
OVRPTW	0.120503	0.193603	0.071996	0.015826
OVRPB	0.122174	0.192740	0.071989	0.0180618
OVRPL	0.120301	0.193039	0.071308	0.0144582
VRPBL	0.120240	0.191676	0.071387	0.0103952
VRPBTW	0.120442	0.191798	0.071435	0.0118685
VRPLTW	0.120597	0.191612	0.071257	0.0205909
OVRPBL	0.120477	0.193109	0.072051	0.0484095
OVRPBTW	0.120987	0.192056	0.071467	0.0210505
OVRPLTW	0.119832	0.193401	0.071603	0.0203865
VRPBLTW	0.119389	0.190899	0.070839	0.0159762
OVRPBLTW	0.120135	0.191298	0.071107	0.0157114

Feature preparation stage wall time: TRAIN/VAL 388.815s; TEST 29.825s (cache reuse allowed).

Original inputs already on GPU; online raw path has no relation_nodes/relation_node_mask.

Transform includes FP16 quantization. All18 tasks include ATSP; no solver decoder.

Excludes dataset I/O and host transfer. Cache-hit selector is not full inference or I/O-inclusive E2E.

Finite64 residuals are recorded, not proof of exact Sinkhorn convergence. Device: NVIDIA GeForce RTX 3090

R99 B vs A | Timing recovery numerical boundary

Private validator revision; approximate messages are NEVER declared bitwise equivalent.

A: Unchanged locked weights/decision rule. Measured path=mixed, TF32 policy=default. Bitwise FP16 equality on first128 rows/task=False; original mixed-online equivalence=False. Full TEST efficacy comes from untouched cached predictions, not a demonstrated exact online run.

B: Unchanged locked weights/decision rule. Measured path=mixed, TF32 policy=default. Bitwise FP16 equality on first128 rows/task=False; original mixed-online equivalence=False. Full TEST efficacy comes from untouched cached predictions, not a demonstrated exact online run.

TEST efficacy: original locked cached predictions only. No TRAIN/TEST rerun or model selection.

LIB: unchanged original reporter, same checkpoint; its native online computation is disclosed separately.